The Kasami Equation-(1) LEGO

Building Dobbertin's Theorem 1, step (1) ⇒ (2) ⇒ cases, from the bricks L and F

A picture book of the `Equation1` Lean formalisation (Lean / Mathlib)

Scope. This booklet describes *only* what the `Equation1/` formalisation actually contains — the equation-(1) slice of the proof of Dobbertin's Theorem 1, together with the trace-free Theorem 5 it reuses. It does *not* cover APN, the MCM bridge (Thm 3/4), the explicit inverse (Thm 6), Theorem 8, or the difference-set corollaries/conjecture: none of that material is in this extract. Every box marked ✓ is `sorry`-free in Lean, resting only on `propext`, `Classical.choice`, `Quot.sound`.

0. The bricks (what is genuinely a building block)

L linearized trace

$$\mathrm{Tr}(x) = \sum_{i<n} x^{2^i}, \quad L_m(x) = \sum_{i<m} x^{2^i}$$

additive: $L_m(a+b) = L_m(a) + L_m(b)$

F Frobenius

$x \mapsto x^2$ (on exponents $e \mapsto 2e$)

cycles, bijective, periodic mod n

C coprime / inverse-mod

$\gcd(2^k-1, 2^n-1) = 1$

$(2^k-1)^{-1} \pmod{2^n-1}$ exists

built from

The identity that glues L to F (Artin–Schreier telescope), ✓:

$$L_m(x)^2 + L_m(x) = x^{2^m} + x$$

(`truncTrace_sq_add_self`, `FiniteFieldPrereqs.lean`)

Brick L — linearized trace.

```
def Tr (n : ℕ) (x : L) : L := ∑ i ∈ range n, x^(2^i)
def truncTrace (m : ℕ) (x : F) : F := ∑ i ∈ range m, x^(2^i)
lemma truncTrace_add :
    truncTrace m (x + y) = truncTrace m x + truncTrace m y
lemma truncTrace_sq_add_self :
    truncTrace m x ^ 2 + truncTrace m x = x^(2^m) + x
```

Paper. "Tr denotes the trace function from $\mathbb{F}_{2^n}$ onto $\mathbb{F}_2$" (§2). The telescope is the mechanism behind "adding the 2^k-th power of this equation to itself" in the proof of Thm 1.

Brick F — Frobenius / doubling.

```
lemma frob_cycle (x : F) : x ^ Fintype.card F = x
lemma frob_mod (hn : Fintype.card F = p^n) : x^(p^r) = x^(p^(r % n))
lemma frob_bijective (r : ℕ) : Bijective (fun x => x^(p^r))
lemma pow_field_bijective (ha : Coprime (card F - 1) a) :
    Bijective (fun x => x^a)
```

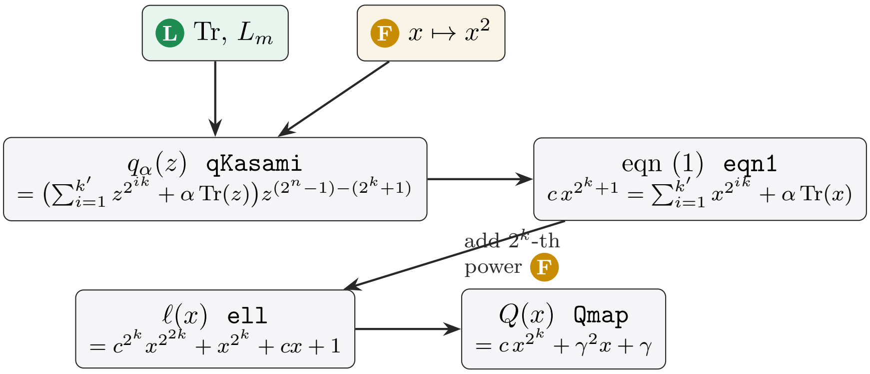
Paper. the doubling of exponents 2^{ik} and the reductions modulo $2^n - 1$ used silently throughout the proof; `pow_field_bijective` is what makes $x \mapsto x^d$ a permutation when $\gcd(d, 2^n - 1) = 1$.

Brick C — coprime / inverse mod $2^n - 1$.

```
theorem mersenne_coprime (h : Coprime k n) :
    Coprime (2^k - 1) (2^n - 1)
theorem inv_mod_exists (h : Coprime k n) (hn : 1 < 2^n - 1) :
    ∃ b, (2^k - 1) * b % (2^n - 1) = 1
```

Paper. "First note that $(2^k-1)^{-1} \pmod{2^n-1}$ exists, since $\gcd(k,n)=1$." In this extract C is present *only* as this exponent/inverse arithmetic (via $\gcd(2^k-1, 2^n-1) = 2^{\gcd(k,n)}-1$), not as the cyclotomic-coset / difference-set combinatorics of the wider paper.

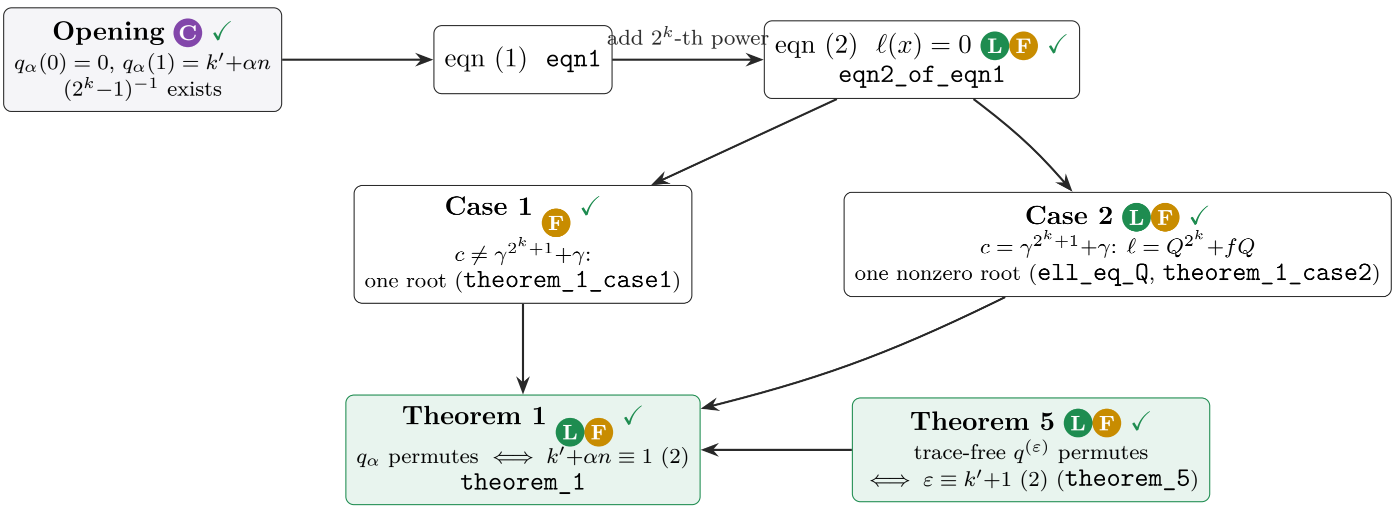
1. The assembled model: definitions built from the bricks




```
def qKasami (n k k' α : ℕ) (z : L) : L :=
  ((∑ i ∈ Icc 1 k', z^(2^(i*k))) + (α:L) * Tr n z)
    * z^(2^n - 1 - (2^k+1))
def eqn1 (n k k' α : ℕ) (c x : L) : Prop :=
  c * x^(2^k+1) = (∑ i ∈ Icc 1 k', x^(2^(i*k))) + (α:L) * Tr n x
def ell  (k : ℕ) (c x : L) : L := c^(2^k)*x^(2^(2*k)) + x^(2^k) + c*x + 1
def ell0 (k : ℕ) (c x : L) : L := ell k c x + 1
def Qmap (k : ℕ) (c γ x : L) : L := c*x^(2^k) + γ^2*x + γ
```

Paper. q_α is the "generalized Kasami polynomial" of §2 (with $1/z^{2^k+1}$ realised as $z^{(2^n-1)-(2^k+1)}$, convention $0/0=0$); `eqn1` is equation (1); `ell` is $\ell(x)$ of equation (2); `ell0` its homogeneous part $\ell_0 = \ell + 1$; `Qmap` is $Q(x) = cx^{2^k} + \gamma^2 x + \gamma$ of Case 2.

2. Snapping the bricks together: (1) ⇒ (2) ⇒ cases ⇒ Thm 1



L linearized trace   F Frobenius   C coprime/inverse-mod   ✓ sorry-free in Lean

2.0 Opening (before equations (1)/(2))

```
theorem qKasami_zero : qKasami n k k' α 0 = 0
theorem Tr_one : Tr n (1 : L) = (n : L)
theorem qKasami_one : qKasami n k k' α 1 = ((k' + α*n : ℕ) : L)
theorem qKasami_one_eq_zero_iff :
    qKasami n k k' α 1 = 0 ↔ (k' + α*n) % 2 = 0
```

Paper. "We have $q_\alpha(0) = 0$, using the convention '$0/0 = 0$'. To verify the 'only if' part, observe that $k' + \alpha n \equiv 0 \pmod 2$ is equivalent to $q_\alpha(1) = k' \cdot 1 + \alpha \cdot \mathrm{Tr}(1) = 0$." (Uses brick C for the inverse and brick L to evaluate $\mathrm{Tr}(1) = n$.)

2.1 The step (1) ⇒ (2): the telescope in action

```
theorem eqn2_of_eqn1 (hn : Fintype.card L = 2^n) (hkk1 : k*k' % n = 1)
    (α : ℕ) (c x : L) (hx : x ≠ 0) (h : eqn1 n k k' α c x) :
    ell k c x = 0
```

Paper. "Adding the 2^k-th power of this equation to itself we get . . . and consequently $\ell(x) = c^{2^k} x^{2^{2k}} + x^{2^k} + cx + 1 = 0$" – equation (2). This is exactly F (raise to 2^k, i.e. apply Frobenius k times) fed into L (the additive telescope `truncTrace_sq_add_self`).

Correction (documented in the file). The bare skeleton is false at $x = 0$: `eqn1` holds vacuously there ($0 = 0$) while $\ell(0) = 1 \neq 0$. The faithful statement adds $x \neq 0$ and the field hypotheses.

2.2 Case 1 — c not of the form $\gamma^{2^k+1} + \gamma$

```
theorem theorem_1_case1 (hn : Fintype.card L = 2^n) (hk0 : 0 < k) (hkn : k < n)
    (c : L) (hc : ∀ γ : L, c ≠ γ^(2^k+1) + γ) :
    {x : L | ell k c x = 0}.ncard = 1
```

Paper. "the homogeneous part $\ell_0(x) = \ell(x)+1$ of (2) has no non-zero solution . . . Hence (2) has precisely one solution, and we are done." The Lean proof makes the linear map $\varphi(x) = c^{2^k} x^{2^{2k}} + x^{2^k} + cx$ injective (additivity is brick L/F), hence bijective on a finite field, so $\varphi(x) = 1$ has a unique solution.

2.3 Case 2 — $c = \gamma^{2^k+1} + \gamma$

```
theorem ell_eq_Q (k : ℕ) (c γ x : L) (hγ : γ ≠ 0)
    (hc : c = γ^(2^k+1) + γ) :
    ell k c x = Qmap k c γ x ^ (2^k)
      + (γ^(2^k-1) + γ^-1) * Qmap k c γ x
theorem theorem_1_case2 (hn : Fintype.card L = 2^n) ... (hc0 : c ≠ 0)
    (hc : c = γ^(2^k+1) + γ) :
    {x : L | x ≠ 0 ∧ eqn1 n k k' α c x}.ncard = 1
```

Paper. "Setting $Q(x) = cx^{2^k} + \gamma^2 x + \gamma$, $f = \gamma^{2^k-1} + 1/\gamma$ we have $\ell(x) = Q(x)^{2^k} + fQ(x)$ . . . precisely one of the solutions . . . is a solution of (1)."

Correction (documented in the file). $x = 0$ always satisfies the cleared `eqn1`, so the faithful count is of *nonzero* solutions (needs $c \neq 0$).

2.4 The headline: Theorem 1 (and its trace-free twin, Theorem 5)

```
theorem theorem_1 (hn : Fintype.card L = 2^n) (hk : k < n) (hcop : Coprime k n)
    (hk' : k*k' % n = 1 % n) (hk0 : 0 < k) (hexp : 2^k+1 < 2^n-1)
    (α : ℕ) (hα : α = 0 ∨ α = 1) :
    Bijective (qKasami n k k' α) ↔ (k' + α*n) % 2 = 1
theorem theorem_5 ... :
    Bijective (qeps n k k' ε) ↔ (ε  ≡ k'+1  (mod 2))
```

Paper. "Theorem 1. A generalized Kasami polynomial q_α is a permutation polynomial on L iff $k'+\alpha n \equiv 1 \pmod 2$." and "Theorem 5. $q^{(\varepsilon)}$ is a permutation polynomial on L iff $\varepsilon \equiv k'+1 \pmod 2$. Again the proof of Theorem 1 can be applied." The Lean `theorem_1` routes $\alpha = 0$ through `theorem_5` and $\alpha = 1$ through the trace-general `gmap_bijective_iff`.

3. Supporting bricks actually used (Theorem 5 / trace layer)

These are the intermediate parts, all in the extract, that the assembly above leans on — still nothing but L, F and finite-field counting.

Lean lemma	role	face
`sTrace_eq_pTrace`	$\sum_{i=1}^{k'} x^{2^{ik}} = P(x)^{2^k}$	L F
`pTrace_telescope`	step-2^k additive telescope	L F
`ell_of_eq`	(1)$\Rightarrow$ $\ell = 0$ core (Artin–Schreier)	L F
`ell0_root_imp_image`	nonzero root of $\ell_0 \Rightarrow c = \gamma^{2^k+1}+\gamma$	L F
`Q_factor`	$\ell = Q^{2^k} + fQ$ factorisation	F
`root_count` / `root_count_image`	counts solutions of (1)	F
`qeps_mul_unit` / `qKasami_mul_unit`	clear denominator on units	F C
`trace_frob_shift`	$\mathrm{Tr}(x^{2^k}) = \mathrm{Tr}(x)$	L F
`trace_artin_schreier_zero`	$\mathrm{Tr}(y^{2^k} + y) = 0$	L F
`trace_sq`, `trace_bit`	$\mathrm{Tr}(x) \in \{0,1\}$	L
`gmap_bijective_iff`	trace (α=1) criterion	L F

4. Census — one pattern, from bricks to headline

Result	Bricks	Lean ✓
$(2^k-1)^{-1}$ exists	C	`inv_mod_exists`, `mersenne_coprime`
values at $0, 1$ ("only if")	L C	`qKasami_zero/_one/_one_eq_zero_iff`
(1)$\Rightarrow$(2) telescope	L F	`eqn2_of_eqn1`
Case 1: one root	F	`theorem_1_case1`
Case 2: one nonzero root	L F	`ell_eq_Q`, `theorem_1_case2`
Theorem 1	L F	`theorem_1`
Theorem 5 (trace-free)	L F	`theorem_5`

One sentence. The linearized trace L and Frobenius F, glued by the telescope $L_m(x)^2+L_m(x) = x^{2^m}+x$ and helped by the coprime/inverse-mod brick C, snap together as opening → (1)$\Rightarrow$(2) → Case 1/Case 2 → Theorem 1 (and its trace-free twin Theorem 5) — and *that tower is exactly what is formalised*; the higher structures of the paper (APN, MCM bridge, inverse, Theorem 8, difference sets) are not built from these bricks here.

Sources: `Equation1/` (`Defs`, `Setup`, `FiniteFieldPrereqs`, `Theorem5`, `Theorem8Trace`, `Theorem8C1`, `Q1General`, `Equation1`). Paper: H. Dobbertin, *Kasami Power Functions, Permutation Polynomials and Cyclic Difference Sets* (1999), proof of Theorem 1 and Theorem 5.